ActInf ModelStream 019.1: Active Inference in Discrete State Spaces from First Principles (P. Kenny)

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Hello everyone, welcome. It's January 15th, 2026 and we're in active inference model stream number 19.1 with Patrick Kenny and Thomas Parr and Norsaget as guests discussing active inference in discrete state spaces from first principles. Patrick has a presentation and also we will hear from our guests and read any questions from the live chat. So Patrick to you for the presentation and thank you. Go for it. >> Uh thanks Daniel. Um so my title is active inference in discrete state spaces from first principles. Now let me explain briefly why discrete state spaces. It's because the what I think is a natural formulation of active inference is the path integral formulation and this is very easy to get to grips with in discrete state spaces because you can do it with a hidden markup model as the dynamic model. It's um not so obvious what uh you can do in continuous state spaces with uh continuous time. If you want to do uh the path integral formulation and uh develop it to uh to the point where it might actually be of some use in uh practical applications and when I say from first principles that's a nod if you like to the free energy principle I'm proposing that um uh active inference the problems that need to be addressed can be tackled by standard um methods that have been developed in machine learning. in particular the mean field approximation uh without having to appeal to the um to the free energy principle and uh I'll explain that there are some limitations that this discipline uh imposes but also some uh advantages which uh which acrew from um adhering to uh traditional well understood methods that have been developed uh in machine learning based learning obviously. So um I'll explain then the mean field uh how that uh addresses the fundamental problem in active inference the um uh processing the uh perception action cycle so that um an agent can uh infer its uh future actions from its past history and um essentially by uh predicting its uh future sensations and uh how this leads to an optimization criterion. In fact, it's a KL divergence criterion which is not quite the same as expected free energy. It's very similar but there is a subtle uh difference which uh which I will dwell on a bit. uh and I will explain that there are other natural applications of the mean field approximation in uh active inference which um do not appeal to the um free energy principle. In particular the uh the problem of basian learning of HMM is a direct application of the mean field approximation. uh also the problem of um of uh updating beliefs about uh policies uh that too is amanable to treatment by the meanfield approximation and then I'll conclude with u the pros and cons of of doing things this way. So here's the the mean field approximation in its um uh most compact form. You have a target distribution which you want to approximate but uh is typically uh intractable in the sense that the well say that the the partition function can't be uh evaluated. You can't normalize it as a probability distribution. That's the typical scenario and you want to approximate it by a tractable distribution which is traditionally denoted by Q and the criterion is the uh divergence from Q to P. This is the form of the divergence which is uh capable of being evaluated under these assumptions that Q is tractable and P is intractable. And the uh the basic assumption is very simple. You assume that uh that Q uh factorizes in this way so that the state space can be split up into streams which are statistically independent under the approximating distribution not necessarily under the original distribution. And there's a very simple rule for um uh estimating or calculating the factors in this factorization for Q. It excuse me it requires that you update these factors asynchronously one at a time according to this prescription. Okay. You take the the log of the target you average out all of the um components other than the component of interest qx of n um by using the current uh estimates of the factors and then you renormalize the result by passing it through a uh sigmoid function in order to get a um a probability distribution. So the things to note about this um first is that the criterion the divergence is guaranteed to decrease on uh each iteration and secondly that you don't actually have any choice in how the update formulas are to be derived. They are determined by the form of the generative model that you're approximating P of X and the factorization that you're going to impose on the approximating distribution. That's different from the free energy principle. You don't have any choice in how you're going to um do the updating. uh it's mo the mean field approximation is mostly used in calculating sterior distributions and in that special case uh the way you handle it is by uh clamping uh one of the um variable substreams to the observed value. you're calculating the posterior distribution of all of the variables other than the v other than the variable that has been observed. And in the mean field approximation, you simply clamp the um that component to the observed value which I've denoted here by x0 bar. Now the main point I want to make in this paper is that uh although the uh meanfield approximation is traditionally used in calculating approximate posterior distributions which of course is the problem of perceptual in inference. It can also be used to approximate predictive distributions so that uh an agent can use it to predict his future sensations and thereby infer the uh actions which he is going to embark on. So um the for this I need now to introduce some more notation. Uh the variable stream I assume is going to be split between hidden variables and observable variables. The hidden variables I denote by S and the observable variables by by O. And the relationship between variation of free energy and the mean field is summarized by that uh boxed equation. This is not one of the usual ways of uh expressing variation free energy. It turns out to be the one that's most convenient for uh for my purposes. If you uh impose the constraint on the factorization that I just um described in the last slide in this notation, okay, where I'm distinguishing between hidden and observable variables, the variation free energy is just the KL divergence between the target distribution and the approximate distribution where the approximate distribution is required to satisfy this constraint here. So uh this uh this is not a one of the usual ways of characterizing variation of free energy but it's a simple um calculation which is um easy to to derive it's it's done in the paper. I'm not going to go into the details. So, so this is how the mean field approximation solves the problem of perceptual inference where you have uh the agent has uh recorded sensory observations and wants to infer the state of the world or the past history of the world up to the up to the present moment. And the problem I want to focus on is how the mean fail approximation can solve the problem of active inference, which is to say how it can produce appro approximate predictive distributions for the future sensations of of the agent. So here's the um here's the picture. Um I I'm assuming that uh the agent is modeling its world by a hidden markup model. Okay. So that all of the agents knowledge about the world including the repertoire of actions, the repertoire of possible sequences at a time zero and the prediction horizon uh extends to a time capital t. All of the uh s variables are hidden. Their values have to be inferred. But at a given time there will be a history of past observations that uh that have been recorded. So I've denoted these by 01 up to to OT using the uh white background. and the future observations which have to be predicted uh I've used this hash hashing to uh distinguish them. So all the hash nodes indicate um variables that have not yet been observed or have to be predicted. So um let's see the problem of perceptual inference is just a matter of approximating a posterior distribution on the states up to the present moment given the observations up to the present moment. Whereas active inference is the problem of it can be viewed as the problem of uh approximating the posterior distribution on the future states and future observations given the observations up to the present and uh what I'm proposing in the paper that uh both of these problems can be solved in an integrated way by minimizing a single divergence between an approximating distribution Q and the HMN P uh where the um mean field approximation uh implements the the mechanics. the um constraints on on Q okay are simply that okay so I'm using the standard conventions here uh Roman T little t is the present time the Greek uh towel is any time in the past and the future up to the planning uh horizon which is capital T. So the uh we have a standard uh factorization where the constraints on the factorization are that um the observable variables their values have already been observed at the present time t and the future uh observables are distributed according to the emission probab probabilities of the hidden markup model. That's the only constraint that's imposed on the factorization that feeds into the the main field uh assuming that the um that the hmm that the state space is sufficiently small that you can pre-ompile the um the matrix of uh transition probabilities. You can work out the um the mean field updates in a fairly straightforward way. the the the mechanics are presented in the paper. And if you're uh in the situation of this is the time index that refers to any time in the past or the future. If you're in the situation where that time index is referring on to the past so that you've already got an observation history, then these are the update formulas. And um I expect everybody watching this video will recognize that these are just standard update formulas for minimizing variation of free energy such as are presented in the uh active inference textbook. The only difference is that for uh a technical reason that uh that I will explain the uh uh Q at uh time towel equal to zero is uh not being updated. It's assumed to be inherited from the uh agents uh prior uh actions. Um so this thing um would be I guess the D matrix in the in the in the standard convention but um there's a slight departure from the um from the standard treatment of time to equal to zero uh in the paper which I will explain uh when I come to it. Now this is the update formula in the case where the Greek uh letter towel the the time at that time index is in the past but and that solves the problem of perceptual inference but for active inference you need to predict the future rather than retradict the past. So there's another set of uh of update equations which uh is almost the same. This uh this h here is the vector of entropies of the emission uh distributions in the hmm. Okay, you can't actually you don't actually have observations for future times. you have to uh average over all possibilities and what you get is just the negative entropy. The um the only difference between this and the previous slide is that uh for times in the past the uh observation history is available. Uh for times in the future you just have to um excuse me for times in the future. this one here uh you just have to average over all possible futures and you just end up substituting this uh entropy uh vector in it in the place of the observations in the past. So these um these uh update formulas both in the past and the future are guaranteed to uh to decrease the divergence between the approximating distribution and the and the hidden market model. So the basic posit here is that if an agent models its world by a hidden mark of model and it wants to calculate a predictive distribution for its future observations, then the right way to um evaluate the quality of the its predictive distribution is to measure its divergence from the HMM. The hmm is what the agent would use if the state space were sufficiently small to do uh exact uh calculations. Of course, our uh basic assumption is that the safe space is so complicated that variational methods are are needed. And the natural approximation, at least to my way of thinking, the natural criterion for deciding how well a predictive is fit for purpose is to compare it with the HMM to calculate the divergence, which excuse me, to calculate the HMM, which holds all of the agents knowledge about the world that inhabits. Okay. To see then what all of this has to do with um the free energy principle. There's a naturally if you look at the constraints, it's obvious that there's going to be uh a decomposition in the divergence criterion, a contribution from the past and a contribution from the future. And you can tease those out. The as far as the past is concerned, the observable variables are clamped to the actual values of the observations. And as I pointed out at the beginning, clamping uh some of the variables enables you to interpret the chaos divergence as a variation of free energy. That's the first component here. the second component. Okay, when you work it out, it's the divergence between the um the approximating distribution in the future and the HMN in the future. But if you look at the constraints that I imposed, okay, I assumed that the approximating distribution inherits the emission probabilities from the HMM. So when you're calculating a divergence, the only possible scope for a divergence between the uh approximating distribution and the H&M is in the state space. Okay, so the joint divergence here simplifies to a marginal divergence and that's this second term here. So the basic claim that I'm making in the paper is that this thing is what is most natural to look at in place of the uh future free expected free energy for the purposes of calculating predictive distributions. Now, as I mentioned at the beginning, there is in fact a very close relationship between this and expected free energy, but they're not quite the same. So, I'll briefly explain what that uh what that difference is. So, here uh in this slide, I've just reviewed the uh standard way of defining expected free energy. Okay. Uh so what we have is if we assume that the posterior distribution okay uh for the future observations the posterior distribution on states or state sequences is independent of the observation. This is something that's done in formulating expected free energy but it's obviously not done in formulating variational free energy. Uh but uh it's has the advantage that you get something that's uh easily computationally tractable. Okay, you start with this uh definition. You do some fairly standard uh manipulations and you get a a marginal divergence which is just like the thing I pointed to in the last slide together with this uh ambiguity term. Then you do some more fiddling and you get the standard uh expected free energy function or uh consisting of the uh pragmatic value and the mutual information or uh information gain term. So that's the standard way of doing things. If you do similar fiddling with the um marginal divergence, uh what you end up with is the uh you get the uh pragmatic value again the same one as here but instead of the mutual information you end up with an entropy term. This is uh can be viewed as an entropy regularizer. Optimizing this will encourage you to um produce approximate distributions that have uh high entropy rather than um uh high mutual information between the um hidden and observables. This I think is a reasonable thing from a biological point of view. If uh if an agent is um in the business of inferring its future actions, then it uh does need to be uh careful not to be overconfident about its uh its future future predictions. It needs to be able to look ahead but also to know uh the limitations the degree uh to which uh its future predictions are to be trusted. So that would be if you like a biological rationale for having this uh entropy term here. Now um this here was um this assumption is can be relaxed. Okay, it's possible to uh evaluate an expected free energy without making this assumption here. And uh when you do that okay you um dropping this assumption variation of free energy if the future observations were available you would calculate it this way and expected free energy you would then average over all possible future observation sequences and you do a bit of um algebraic manipulation and you find again an an entropy term that um relates this generalized expected free energy to the marginal divergence which uh I have suggested is the most natural way most natural criterion for predicting the future. So the relationship between the criterion I'm proposing and um expected free energy uh defined in this generalized way uh is the only difference between them is an entropy regularizer that uh encourages approximate distributions which are of high entropy. So uh I think I can say that what I'm proposing uh is very faithful to the spirit of the free energy principle but there is a subtle difference um that has to do with this uh regularization. Uh there are a few other points I could cover but uh I'm not going to linger on them. just um to show firstly that the um standard um basian method of learning HMM parameters is in fact although it may not usually be presented this way is a direct consequence of the mean field which is just a general procedure for approximating complex joint distributions by simple factorized distributions. So all you have to do in any application is figure out what is the joint distribution you're trying to approximate and what is the So to the joint distribution in the case of um learning HMM you need a joint distribution on um the HMM parameters and the hidden states and the uh observations which is you see this second term here is just the HMM and this is the directly prior on the HM parameters and then you approximate it uh by imposing uh a suitable factorization and then you just turn the crank. Uh the mean field approximation is just a straightforward mechanical procedure once you have set the thing up properly. And what emerges in this case is just the uh standard procedure for uh estimating hmm parameters. Uh one other application which is a bit more technical but I explain it uh in detail in the paper is how the uh mean field approximation applies to the uh problem of updating beliefs uh about uh policies. Um this uh assuming that you have a prior distribution of policies that would uh I think typically be denoted by a vector E in in the in the active inference literature and you have observations in the past and you have a probability distribution. Well, you want to calculate a posterior distribution given the observations in the past. Um you can do it just by uh figuring out exactly what is the joint distribution of the hidden variables that you're going to approximate. What are the uh conditions that you impose on the factorization in order to make the mean field updates computationally tractable. And you end up with a formula which is very similar to uh an equation in the active inference uh textbook with the difference that what plays the role of expected free energy is this thing this contribution of the future to the kale divergence. Okay. And it's not an expected re-energy. So uh that's a bit too technical to uh dwell on any further. I'll just say that anyone who's interested can find it in the paper. So I can now uh conclude so the the mean field approximation can be brought to bear on all of the basic problems in the uh in active inference. Okay. The three that we looked at was how to infer how an agent can infer a predictive distribution on its future sensory from uh an observation history data that has been collected in the past. Um how um the method approximation applies to the problem of basian learning of hmms. how it um enables you to uh update beliefs about policies in a a um principled way. The limitations of the method are that the objective function has to be a kale the the function that you're trying to optimize. This uh presents no problem in uh the case of perceptual inference. Okay, because the although it's not usually done, variation of free energy can be expressed directly as a kale divergence. However, this is not true of expected free energy. expected free energy. No matter how you define it, it's not a scale divergence, but you can manipulate it uh by adding in an entropy regularizer so that you actually do get a GL divergence so that the objective function is amanable to the mean field And the critical point I think is that the uh update formulas for the mean and field approximation are determined. You don't have any choice. They're determined by the joint distribution of the variables you are uh which are germanine to the situation that you're working in and the constraints that are imposed by the factorization. Once those have been written down, everything else was mechanical. But you don't have any choice in writing down the update formulas. Everything is governed by the generative model. That's my sort of basic assumption that an agent that's navigating a discrete state space using a uh HMM to encode all of its knowledge about its world and that governs the um the updates. The major advantage I think is that there is a convergence guarantee. the uh if you uh have implemented things properly, the KL divergence is guaranteed to decrease on every belief update. And as far as I know, there are no convergence guarantees for the free energy principle. So that concludes my talk. Um, I guess I'll continue to um to share the screen in case there are any questions about um uh any formulas or anything. >> Thank you. Great. All right. Yes, leave the slides up, Thomas, if you would like to make any first comments or I can just ask some preliminary questions. And meanwhile, people who are watching live can write in the chat. If you'd like to go first though, Thomas, go for it. >> I'd suggest we do it that way actually, Daniel, because I'm interested in what people ask and then we can sort of discuss some of those points afterwards if that's okay. >> Great. Okay. So, uh, people watching live can write in the chat. Let me just ask a few clarifying questions while people are writing, Patrick. So in the first introduction of the VFE, >> if you could go back to that slide on I think decomposing the divergence. >> Oh, the the slide, excuse me, the slide entitled decomposing the divergence >> a little bit back >> there. Okay. >> Uh when I mentioned the variation of free energy was here. >> Yeah. Yeah. Okay. Thank you. Here. So here you have the VF defined by only the KL divergence between the joint distribution of the Q approximation and the P. In equation 2.5 of the 2022 textbook in the third line, there's a divergence and an evidence term. So have you just only focused on the KL divergence aspect or is is there a different role for how you've derived? >> This this is literally the same as the um standard presentation. There are actually three standard presentations of the variational free energy and what I'm saying is that this one is uh exactly the same just written in a different way is it it actually requires a little bit of pencil and paper calculation to see that but uh it's it's in the paper. >> Okay. Perhaps just to make to make that clear. So you know that that the um kale divergence here can be separated out into a difference between um an entropy and an energylike term just as the the variational free energy would be and the the entropy term comes from the first the cq bit here. Now if you were to separate out the but because the entropy is a log term you can separate out into a sum of the entropy of Q of S and the the um delta distribution here. So the entropy of the delta distribution is going to be zero by definition. So will vanish which leaves you with just the Q of S bit. So it's exactly the same. >> Okay. It is obvious to you then. Good. It Yeah. Good. >> Okay. So it wasn't it wasn't obvious to me. I have to actually uh if you look at the paper I I derived it but uh in a more securious um way. >> Yeah those those different decompositions of the VFE in equation 2.5 and the EF which what my next question will sort of be about. It's very fascinating these kind of rings of compatible isomorphic or transformable equations that all highlight different aspects. And when it comes to the computational implementations, sometimes certain representations have different computational uh availabilities or resource needs depending on on state uh space size and all of that. >> So just >> indeed. >> Yeah. No, I would just say that for my purposes, okay, it was essential to cast the variation free energy as a single scale divergence and nothing else because that um that's what the mean field approximation uh is tailored to optimizing. So it just reinforces your point. There are many different ways of writing uh these functionals which are u more or less suitable depending on the um context that you're working under. >> Great. And then just one kind of similar clarifying question about the >> EF >> that was a few slides later. >> Okay. So just >> so I had a couple of um here perhaps or >> or here. >> Yeah, >> this was the preamble to the um developing the relationship between this divergence criterion and the >> but the um >> yes go. Yeah. So, so just on on here sort of conventionally the EF is modeled as a sort of tree search unrolling of conditional action sequences over the policy time horizon and the consequences of the the transition matrix slice constituting the B tensor. So ho how does the mean field or your formalism here deal with the conditional effects of different actions? How can we average over expected observations in a way that at least here though I see later the pi variable and the action conditionality comes into play here I was a little surprised to see this as a sort of temporal unfolding Gaussian blur over observations that doesn't immediately seem to condition upon action selection. >> Okay. Uh good question. the um in fact it's it's really comes at the at the heart of the matter for for this uh for this paper. My um assumption is that the uh agent encodes all of its knowledge about its worm in a richly structured hidden market model. So the the type of model that I have in mind is the uh the sort of model in the pixel what is it called planning from pixels paper the the one which uh gariston has called the reormalization uh the renormalization group model I I forget exactly what his terminology is but where where you have a a hierarchical structure Okay. Where the uh actions and policies can be folded into the uh into the way that the HMM states are defined.

Okay. So there's there's you actually although the expect the the free energy principle is usually presented as a way of um of uh making inferences about actions which are currently underway or are yet to be performed. That way of presenting things uh hinges on a hard distinction between actions and uh and states. But if you look at the sort of hierarchical uh structure in the um the um what's that paper called? Planning from pixels. I I forget what >> pixels to planning. >> Pixels to planning. If you look at that hierarchical structure where actions are just transition probabilities on states, okay, and the state structure then is redefined so as to uh incorporate paths and states into a similar into a single structure. So you no longer actually have to talk explicitly about actions. Okay? everything can be um formulated as inference in terms of a richly structured uh hidden market model. Well, that that's the perspective that I'm taking in the paper that uh a sufficiently richly structured model encodes world, its repertoire of actions, its repertoire of uh policies and it enables the agent to do the um path integral calculations. or um in order to make its uh predictions. >> Thomas, do you want to add any comments to that or I can read two of the questions in the chat? >> Um I I think I'll have more comments on and questions about this slide later, but let's should we do the questions first and then we can come back to that >> Great. All right. Alexander Sabine in the chat wrote, "Is there any sense in which active inference tracks the cumulative cost of these KL updates over the long time horizon?" >> Well, the KL divergence here is not the divergence from one static distribution to another. The the Q uh the approximating distribution is approximating a hmm. So both the Q distribution and the P distribution are dynamic models. So it's sort of it's built into the KO divergence as long as you recognize that it's the divergence from one dynamic model to another rather than the to another static distribution. >> Thank you. And that to me at least sort of rhymes with Rand Wei's work on reinforcement learning and active inference where the pragmatic value is reflected by a MDP and then there's a partially observable element which contributes an epistemic uncertainty and it's that double superposition of the the pragmatic preference constraint oriented and then the sheer entropic uncertainty which together in a joint functional make it active inference. or indeed if I sorry but with it reminds me of something I I I really should have mentioned that there there is this uh paper by um Hafner uh whose title if I remember correctly is um is it action and perception as divergence minimization something like have which uh is sort of unifies um many many different uh algorithms in um uh reinforcement learning under that single umbrella of divergence minimization and uh this too fits under that uh umbrella as long as you understand that the divergence that I'm talking about here is a divergence between two uh dynamic models. Okay. Rather than a divergence between two uh static distributions. >> Great. Okay. And then second question in the chat from Bert Burkers wrote, "Could you explain H and the update rules in more detail?" >> Well, there um there's a sort of Sorry, just a sec. Um the there is this um general perfectly general mean field approximation which can be used to uh approximate any probability distribution by a suitably factorized approximate distribution on the same variables. you simply by fiat impose statistical uh independence assumptions between the variables. That's what this means. And um you try to find the uh distribution that best approximates the distribution you started with in the sense of K's divergence subject to these statistical independence constraints. And that's a that's a universal um mechanism. And all I've done in this paper is apply that to uh in the case of hidden markup models. Okay, where um the distribution is not a static distribution but a distribution that unfolds over time. And the approximation, the constraints on the approximation are that well the mean field constraint which says that the uh variables at uh different times are statistically independent in the approximation. That's just factorization. And the other constraint is that if you have an observation history at your disposal, you clamp the observable variables to those observed values in the past. Okay? In the future, the future is wide open. You just use the uh emission probabilities from the HMM. And then once you impose that, everything is in principle mechanical. the uh the update formulas uh are just a matter of turning the crank. Uh I've I've written them out in detail in the paper that uh accompanies this uh this presentation. Uh it's no surprise that in respect of times in the past, the update formulas are just going to look like the standard uh variation free energy updates. and the um updates in the future look uh very very similar there. There's really not that much uh difference between them. >> Can I on that one? >> Yeah, Thomas please. So in so the the way in which the updates from a mean field approximation are obtained is that or one way in which one can find those formula is that if one takes the gradient of the kale divergence or the functional derivatives and finds the minima uh that then um the minima is what you end up with there is the the update. So they are the fixed points uh so or the minima of that objective function. Now I was puzzled by the hes here as well. And the reason I was a bit puzzled is because the kale divergence you've written down here with the factorization for future states should essentially and you show this on a later slide should cancel out both of the O's because you will both the Q of S given S and O and the P of S and O. You'll be able to factoriize out a P of O given S from both of those. And because the kale divergence deals with the difference between uh two log of probabilities, those all completely cancel out. And as you show in one of the later slides, this kale divergence is exactly equivalent to Q of S P of S with no O's involved in it at all. >> That's right. >> So for the to end up with the um emission probabilities in the um in the update formula seems to me inexplicable. Okay, >> because they should cancel out and then once one takes the the derivatives and finds them in the kale divergence, they shouldn't be there because they're not in the kale divergence to begin with. >> I I I agree they're not in the kale divergence, but the aim of the um of the mean field approximation is to minimize the scale of diversions. Okay. What you want what you want to do is update the um uh um factors. Okay. So as to arrive at at least a local minimum of the kale diversions. And for that uh you do need to take into account the fact that the um emission probabilities the the state the states in the HMN are um as they say heteroscadastic they they have different um entropies. If the if all the entropies were the same, okay, what would happen is that the hes would effectively drop out because uh what the what's going on here is that to get these factors, you pass the right hand side through a uh sigmoid function. Okay? And if the all of the components of this vector H were the same, if all of the HM states had the same entropies, then indeed the um the H's would disappear. They would not play any role. Um does that >> I'm not trying to follow where you got the hes from in the first place. >> Okay. Okay. It's simply the um the it's a vector having one component for each state in the hidden market model and the value of that component is the entropy of the emission probability distribution for that state. >> Yes. So if you just look if you just if you just look here I think you might see what I'm getting at. >> Sorry. Do you mind if I just clarify by probability you you mean the probability of B of O given S which stand to be emission probability as well. That's right. What I'm saying is that that emission probability doesn't exist in the kale divergence because it cancels out. >> I agree. I I agree it doesn't. But the um the mean field approximation and this is just a mechanical uh um derivation okay does require that you take account of the entropies of the HMM states in updating the making predictions for the future because obviously uh the visiting uh visiting states with uh with high entropy is going to have an effect on the value of the KL diversions that you that you are optimizing. If I could just point to one thing which I think might actually clear up the confusion. uh if you look in the what's going on here this a matrix is the what I'm calling the emission probabilities okay so this is how the emission probabilities enter into the standard variational free energy updates right more or less taken from the tech from the textbook I mean I I try to align the notation with the uh with the text file. Okay, that's in respect of observations in the past. Now in respect of uh observations that have yet to be collected okay what becomes of this term okay is just the expected value the conditional expectation of the observation given the state or the log of the um probability given the state and that's how the entropy emerges. in in this uh when you're predicting the future. >> Okay, I don't want to spend too long on this point, but I I think you know for me that that's not entirely clear still and I I appreciate that the emission probabilities appear in the previous version. Oh, sorry. In in relation to past states and that's absolutely the case because the kale divergence in relation to past states is a functional of the emission probabilities. If the kale divergence of the future states is not a functional of then in in finding its minimum if that if that is what we doing to derive what we're describing as the mean field updates then the the minimum is invariant to the emission probabilities if you see what I mean I I I sort of appreciate the the sort of formal similarity um but it feels like that probably the issue comes as to how you're defining your mean field updates and where they come from and if they come from minimization of the kale divergence they can only depend upon things that are in the kale divergence. uh the kale divergence simp okay I think the only answer I can give you at this point okay given that we're sort of going around in circles is that the derivation of this is a simple mechanical consequence of the mean field uh update formulas and if you look at the paper you find that I I stuff these out in painful detail in uh in an appendix because I didn't expect that anyone would actually want to uh grapple with them. But since it appears that that you might want to do so, do by all means and if it's still uh unclear, I can straighten it out. >> Let's leave that point there then, shall we? as going around in circles. Daniel, are there further questions or or either from you or the chat >> here? I was just wondering about the shape of these. If H is already vectorzed, then we're calculating ln Q kind of going laterally through time. But then I just don't see how I I like I I also will will like to think more about it. It seems like one of the key moves is to make action of um make of comparable kind exter so-called external or internal causal forces so that one's own actions are part of the latent states. So all causal unfoldings in the future can be treated in a common setting. And then the question that arises to me is where does the mean field have where where does that approximation and a KL updating lead to model adequacy? Because even if retradiction and prediction updating are guaranteed to decrease through the divergence, is there any way to diagnose our absolute adequacy or do we need another kind of benchmark otherwise the KL divergence convex optimization may not reflect destruct because it's conditioned upon the factorization of the generative model. It may have limited applicability in practice even if it has well well-behaved update rules. >> Yeah. My my assumption the the basic assumption perhaps I didn't make it clear um sufficiently is that the uh agents all of the agents knowledge of the world can be encoded in a s in a hmm of sufficiently rich structure and if that is given then the natural criterion for uh an agent calculating predictive is just the degree to which the predictive distributions differ from the predictions that would be made by the HMM and that is encapsulated in the So the in the way that I am thinking of things the generative model takes center stage and that's not the way that the free energy principle is usually presented and I think the reason for that is that in the um original continuous state space continuous time formulation of the free energy principle the um generative model was sort of a stub the there was no uh obvious candidate some sort of stochcastic differential equation was to play the role of the generative model. Whereas in the uh discrete state space uh setup there is an obvious well understood uh generative model to use namely a discrete HM at least a discrete state space hmm and if you put that in the center then and you say that that is going to rule over everything Then the tail divergence I think becomes the natural criteria for evaluating the quality of predictive distributions. How well do the predictions approximate the predictions that would be made by exact calculations with the H&M? Yes, Thomas, if you have any questions or I'm curious how this lands or how you see this in light of the high road and low road approach that structures the 2022 textbook. >> Yes. Um I mean that there's a lot that that we could talk about here and um I guess first of all I should say you know thank you Patrick for for such an engaging presentation. >> Thank you. It was my pleasure to talk to you about this. I I think um you know it was a really sort of clear and wellresented um introduction to uh first of all use of meanfield methods in um in basin inference but formulated in terms of kale divergence minimization and I think also in terms of that notion of of prediction and inference about states that have yet to happen and and we yet have yet to have data about. Um and I think it's really interesting thinking about the relationship between that and expected free energy and and a number of other things. Um there are a few there there are one or two points in fact one of them we've spoken about already another one we'll come on to that that I would take a little bit of issue with in terms of the technical aspects but some of the other points are more a matter of of being completely valid in and I completely endorse what you say about the the right sort of predictive distribution is effectively the distribution that aligns with the prior. I completely endorse what you say that the generative model is the thing that should always be center stage and should be the thing that uh that determines behavior of any sort and I I I think that's at the heart of the active inference approach. Um uh and um apologies if we weren't clear on that in other in other areas. Um so I suppose first question I have really is what is going to be most useful for you? What what what did you want to get out of sort of initially contacting me about this paper and what did you want to get out of writing it? I suppose >> well it started off reading your book. I I I h I had a um I happen to have a a background in in basian methods. uh um I'm retired but I I used to be a research engineer in in specifically um speech recognition and speaker recognition and um that was my hobby uh developing basing methods to do that. Um I saw basing methods pretty much blown completely out of the water with the uh uh you know with the deep learning uh uh success you know the transformer and so forth with one exception which was the uh the um active inference uh community. So I made it my business to um to read the book and and had a lot of trouble with it and ended up rethinking it uh in the light of my previous experience with uh with the mean field approximation. this it's something that I happen to have used all the time because of the guaranteed uh convergence um that's that's built in so that you have a way of debugging uh complex variational algorithms. You just check that on every belief update the um uh often as the divergence criterion uh decreases and as far as I can see there is no such uh guarantee available for the free energy principle. Um it's plausible that uh that it should that the belief updates should uh decrease the optimization the optimization criteria but I I I don't see any way towards a convergence any sort of convergence guarantee for the free >> That's a really interesting point and perhaps that's a an interesting um place to start. So I suppose from some aspects there is the the sort of exactly the same guarantees as you have for the mean field approach you've described in that the variational energy is exactly the kale divergence you've talked about and uh the fixed points of the kale divergence are also going to be the fixed points of the corresponding variational free energy. Um so in that sense any guarantees that hold for one also hold for the other. Um >> could I just interrupt there and say that um expected free energy and variation of free energy >> uh two different things >> they're two different things >> absolutely >> go ahead what the one the one is the kale diversions and the other cannot be expressed diversions >> absolutely right >> okay yeah >> absolutely right now that's an interesting point in itself and um and again what there's A lot we could talk about here and and we may perhaps come back to that. Um the point about convergence though so convergence for the variational free energy and for inference under that as you say is as guaranteed as it is under any other meanfield kale divergence minimization. Um you said a few times that that there is only one way of of uh using the mean field approach. Uh and perhaps that's true under under some definitions and conventions. Uh we would often use it in in the sense of it is this factorization. These are the fixed points we want to reach. But actually you often we would use a gradient descent process as an alternative. I mean the fixed point iteration you describe is absolutely valid. Um but sometimes you can also particularly if you're trying to model biological systems express these things. >> Yeah, absolutely. If you if you have continuous emission distributions, you're almost certainly going to have to do gradient descent rather than the coordinate ascent which um happens to be so simple. I'm sorry. or if you're doing inference in continuous time as well. I mean so the probabilities that indeed I I made a sort of blanket assumption in the paper that I was just going to talk about discrete distributions and discrete time and in that case the why not use the coordinate ascent rather than the uh gradient descent because it's so straightforward. >> Absolutely. No, it's it's a perfectly valid way of doing it and and often faster. Um but uh but it's worth saying that even if you're using a model based upon uh discrete categorical random variables, the parameters of the distributions describing those are still continuous variables. And so uh you can just as easily perform a continuous coordinate descent. And there may be differences between the times that are being described by the model and the time over which you're doing the inference um within each of those steps. So you know that that becomes much more when you look at biological >> well of course it's also true for coordinate descent or the other one the the what's the performance is critically dependent on the starting point. you're only ever driving towards a local minimum of the of the optimand. So, um you're never you're never going to find a perfect solution to the uh to the problem and the step size may indeed uh affect the outcome of the um optimization. >> Yes. And but then just to come back to your point about convergence and expected free energy and I suppose that the question is do you think that biological behavior and cognition and exploration and decision-m is convergent? Isn't some would argue that there isn't open-endedness to the way in which we behave that it actually doesn't >> clearly the the functioning of the brain is riddled with um closures and and evolutionary hangovers and so on and trying to reverse engineer that is sort of pretty well n a hopeless task. I I um I mean to to exactly uh map onto uh biological function seems to me to be too ambitious. But the the um I'm really looking at this as an engineering problem. that happens to be my I mean a very um unusual challenging engineering problem to actually simulate the behavior of biological uh systems or or subsystems but um I I I I really don't know enough to say whether incilico algorithms could actually translate to too to wet where uh it's really beyond my um um outside of my wheelhouse altogether. >> Well, I suppose the analogy I'm I'm sort of implicitly drawing is that for a biological creature, the only real fixed point they will reach is death. >> And in the long run, we will >> Yes. No, it didn't to answer to answer the question you asked about why why I got into this was was I I I really thought this was tremendously clever idea of that of of uh extending you know the analogy between variational inference, thermodynamics uh perception into into action because you get so much more if you can make that um that leap uh which is basically the transition from variation free energy to expected free energy and that's what u set me on this uh on on this quest to initially just to see if I could understand uh expect free energy and I had so much trouble with that that I found it easier to um to work things out for myself. And this is this is what I came up with. I mean, I I don't have any real stake in the um in the future of this. I I I'm no longer at a stage in my life where I maintain a CV or willing to pay for page charges to uh to publish this sort of thing. It's just the um a hobby that I >> Well, okay. I mean, I think that's useful in terms of, you know, this is an exercise in exploration and curiosity, which I think is is also something that's worth coming back to. um the so but but you've sort of very nicely introduced this question of how do we go from this idea of variational free energy and and the like to the idea of expected free energy or if we're not going to do that what's what's the alternative way of formulating it and actually one of the things I found quite interesting is that um that what you've described is actually very very closely related to the way in which um sort of the the continuous state space formulations of active inference have typically worked where it is all about I have some generative model under which I'm making some predictions a short way into the future and taking actions to fulfill those predictions. Um and that is that is precisely the sort of reflexive like actions to fulfill those predictions uh that the the um the continuous states those models have typically used and actually those don't tend to appeal to expected free energy. That's bit something that's much easier and much more tractable to to relate to um the the the question of discrete state space models as you spoke about before the idea that actually if you can turn a path integral into a sum over time it becomes much more straightforward to evaluate those alternative paths and uh and the temporal aspect of it. Um the way it's typically or or the way one typically addresses paths and predictions into the future in the context of continuous states space models is to make use of generalized coordinates of motion which are effectively a way of writing down a Taylor series like approximation of a local trajectory. And in >> Could I ask you a question about about that? It seems to me that that doesn't uh take into account the the random fluctuations that are in the Livang equation. you you have the epsilon term or whatever you call it. If you simply take a tailaylor series expansion um about the state at a given time, you're not taking account of the um the random shocks that the um that the environment is going to uh inject. So it seems to me to be too cheap a way of trying to get from a dynamic setup to a static setup. If you if you if you represent everything by generalized coordinates, you you simply got a static Well, you assuming you don't update them as you go and and >> Yeah, but but that's that's expos the the the the update if the if the shock comes in but you're not able to anticipate the effect of those uh random shocks if you're use at least this is my naive reading of the continuous state space and I tried to um I tried to make clear in the paper that at least from my point of view the right way to straighten out the conceptual issues in active inference is to start with the discrete state space simply because everything is so much more straightforward and tractable. >> Yes, I mean to try and answer your question but as you say let's stick to the discrete state space on the whole here because that's the subject of your presentation. Um to to answer your question as to how you deal with the random fluctuations, if you have a distribution over both over each element of your generalized coordinates of motion over each of those coordinates, then that will tell you how how confident you are about where you will be at different in time. And so the effect of the random fluctuations comes in through the the the variance of each coordinate in turn, which means that the further you get away from the current point, the greater the uncertainty grows in both directions. And you know, for that reason, there's actually a finite number of generalized coordinates that are typically useful to to employ in a generative model because beyond a certain point, they add no more confidence to the the the um the distribution side. Um, and there's lots of really interesting things one can do looking at the the co-variance between coordinates as well in the context of of um smooth noise processes and and correlated noise processes. >> So it's a fascinating area but but as you say let's stay away from that one for now otherwise we'll end up put down there a very interesting >> and and you know in terms of what you've you've set out I do I I do sort of completely agree with you that that what you're describing is the right way of deriving predictions. about what's going to happen next. And in fact, exactly what one would get if if one um included in a a standard variational free energy future observations as uh as if they were hidden states. One would end up with exactly the same updates. Again, I'm not completely sure about these hes, but we'll we'll leave that one. Mhm. >> Um but now the the sort of discussion about divergences and you know some of the papers you mentioned some of which I've been involved in some of which I haven't um the sort of action and perceptionist divergence minimization then I think in the paper you mentioned Baron Le's paper wz the expected free energy um you know we in a way one of the reasons that this is so interesting or one of the signs that this is so interesting is that these are issues that get revisited every every so often >> and you know I remember having conversations like this nearly 10 years ago now >> sure >> the same sort of arguments and same sort of discussions and I think I mentioned to you in an email it's almost a right of passage if of people as they start >> world >> um and and you know it's it's telling that this is something that captures people's um approach and I think it people's attention and it's one of the really interesting things that to some extent I think is still unsolved in active inference is why is it that something like the expected free energy works as well as it does? Why does it capture as as many things as it does? What's the relationship between it and why does it look so similar to something like the the free energy the variation of free energy without actually being the same sort of quantity in itself without being directly derable from it on that point while we're there the other one point that I I I disagreed with um in terms of what we've actually said was there was a slide where you you defined the expected for energy Daniel went to that slide earlier on >> as I find that I find that just >> Yeah, that would be great. >> This this is the this I presented as the conventional treatment of expected free energy and this as >> what I'm proposing your conventional treatment. >> Okay. So the point I wanted to make for for anyone who who's not used to this is that this is not the conventional treatment. So the the line here that says EF equals the expectation of VF is not the conventional treatment. Um and um I I think there probably are some papers where something like this appears. It certainly doesn't appear in the book. Um and actually has been a point of confusion for quite a long time. And probably part of this is um is that we shouldn't have referred to it as an expected free energy because it is not the expectation of variational free energy. >> Okay. Well, just in in my defense, I I um I have spent a lot of time puzzling over this um thing here. >> Okay. And I wrote this down as what appears to me to be the simplest route to arriving at that and and stuck this in as a flag. I mean the red color as a flag to uh indicate that um there's some threatening going on. >> Yes, >> that's an important point though Thomas. So thank you indeed. There's been the whole EF free energy of the expected future expectation of variational free energy and that was an important point. >> Yes. By the way, Daniel, I should say do interrupt me if I'm talking too much or if there are other questions coming through. Um assuming that there aren't at the moment. Um I think you know we we we we sort of come to one of the key points here which is what what is the expected free energy there to do? Um and where does it come from and where it comes from is a complex question and I think in many ways the way in which this question has been approached has now almost turned on its head that you know this functional form appears to capture a lot of interesting behavior. remind me if I don't come back to it. The other point we we should talk about is um kinds of behavior that might be captured by an expected free energy but would not be captured by the formulation you have here because I think that's another really interesting and important point to come back to. But in terms of the motivation for it, I think that there seem to be a lot of behaviors that are really well captured by it. And it's fascinating that they are captured by something that has such a similar functional form to the variation of brain energy. Um and the reasons for that I think are are to to many in many ways still under investigation and the way in which that's being addressed now particularly in a number of the papers that have looked at the kind of underlying physics of um of the free energy principle almost turn the question round and say given this seems to be such a good description of a system what sort of minimal properties of a dynamical system what sort of conditional dependency structures in the system itself have to be satisfied before this emerges as a natural formulation. >> Well, I I I would um um if I was pursuing that line of investigation, okay, I would ask uh what sort of generative model leads naturally to this particular uh update formula? Is there some way of defining a generative model such that the mean field approximation or something similar actually enables you to derive this? And I think the answer to that is is that actually the way most of these models are derive or defined is that the expected free energy is not an objective function in the classical sense. It is part of one of the priors. It it it is the way in which a prior over what I'm going to do next is defined. And that's the thing that's not present really in in in in the formulation even here. I I I try I did I did try to um satisfy that objection by by deriving uh this um this update formula for policies in the framework that um that I developed. Um the the office I anticipated that um this was going to be an objection and wrote this section in anticipation of that. Whether you're satisfied with it or not, I I I don't know. But I I I I started I mean the the the where I really stumbled trying to understand free energy was or expected free energy was how this equation your your B9 this is my analog of it okay could be derived from this update formula I mean that they just don't seem to go here that was that was >> because it Yeah. So that that's I think exactly the point I'm making that that one does not derive it from this update formula except in to say that if um so if we were dealing with a convergence of Q of pi P of pi then it would depend upon the form of P of pi. And if P of pi is defined in such a way that we expect the actions we take to minimize expected free energy or or to be sort of inversely monotonically proportional to expected free energy then it emerges directly from this biting. >> No prior the objection I was making. Okay. >> Uh so I have to toggle between two slides. So uh so here we both agree this is in my reading the E E vector. Okay the prior unhabits. >> Yeah. >> Okay. This is the variational free energy of a policy calculated with the uh observations up to the present moment. And this term in equation B.9 is expected free energy. And >> that's the point where that's the point where that's not true anymore. >> So from the perspective of the way this equation would be expressed in the active inference book um both the f of the the less than or equal to t and the f of greater than t are both included as a single f term. the expected free energy would actually be part of the log p of pi. >> Oh okay. But um okay so there are two ways you can um implement this type of thing. You can take it that pi is a or the p of pi is um a prior distribution on um policies. Okay. I think the term habit might be used. Okay. And then that can be updated at every time at every time step. >> Yes. So you can use the Q at calculated at time t as a prior at time t + one. >> I sort of left this vague in in formulating this uh as to whether that uh prior in policies was something that was given in advance and never changed or uh was susceptible to being changed as uh observations were accumulated. So it could be implemented either way. So again, >> I think I think I mean I I I would have to I would have to think in terms of >> Sorry. Do you do you finish what you're saying? >> No, I I I I'm winging it at this point. I I I would have to think about it more carefully if there is an issue here. >> Yes. So I'm not saying there's an issue with how you formulated it and what you what you've written is perfectly reasonable for a sort of passive observer. um who or or an observer that behaves in a purely habitual way of working out what I'm going to do next. The that the um and part of that is taking evidence from um what what I've done so far and and that's um what the the sort of ferms here are contributing. They're contributing a an element of uh the evidence in favor of this policy from the things I've already done in the past. Um the the the big difference is that the the assumption that one can derive the expected free energy from this is not correct. Uh and that is not what it's used for. The expected free energy is not something that's used to derive an update scheme. It is a way of defining P of pi um which may have a habitual component and may also have a component from the expected free energy. So one could write it as if you wanted a habitual component you could do that P of pi is a sort of E term as you've said multiplied by the exponent of the negative expected free energy but that is a definition of a prior it it is not a a derived prior >> okay I I'll take your word for it I I may have misread the um the the presentation and misunderstood some um basic point Yeah. >> Well, I I think it it's a point that comes back a lot and you know, I I completely understand why that's a it's an unusual point to get one's head around that if we're so used to seeing an expected free energy or a kale divergence as being the objective function to be optimized. Um it it feels strange to use it as a way of of defining a pri and it's something that doesn't happen in most of basin infants because the idea of defining a prior based upon a function that is itself a functional of uh of our current beliefs and current posteriors feels like a very strange thing to do. But this is one of the the things at the heart of active inference that really makes it differ from things like reinforcement learning. It's the idea that the that a good policy or a good thing to do is not defined just in terms of what I might get at the end, but actually in terms of our beliefs and how those beliefs might be optimized over time. Um and and so what you've written down is not wrong at all in the sense of um in deriving a way of coming up with a good predictive distribution based upon some alternative trajectories I might be on perhaps with a prior bias for thinking I might be more likely on this trajectory than this one. But what it doesn't get at is a form of agency that that uh that sort of updates based upon either explicit goals or or in terms of um in terms of curiosity and belief updating to minimize uncertainty and to resolve uncertainty about the world around us. And I think that's the point I was I was going to come back to about the the difference between these different approaches. Um and you can see that in in in that there is you know what what it comes back exactly to our point about convergence doesn't it that if the uh effective objective function that you end up with is defined as just the kale divergence between two um between a prior belief about how things will play out and my beliefs about how things will play out. then one converges to a point where the H&M matches uh my beliefs about how things will play. Um and so there is a convergence there. Whereas if one is actively selecting alternative trajectories based upon how they will optimize my beliefs in the future, that's a very different sort of of of model, something that has a lot more autonomy to it. Let me let me ask you um the the um at least the the basic implicit assumption that I'm making is that uh all of the um the agents beliefs about the world and uh it's uh the actions open to it and the policies and so on can be encoded in a single HM in princip. principle assuming a sufficiently rich structure and if if that if that encoding is done then uh there is no explicit role played by um policies. >> Yes, I noticed that in in in your paper as well. Um, and I think that's a good summary of of why that's different to sort of more conventional act of inference that that it this is not about policies. This is not about planning. Uh, this is about predicting the future. >> I I I actually agree with that last statement that you made unreserveably. So I I am uh asking then uh if um holding uh actions and policies into states is somehow an impediment to uh making predictions about the >> It depends how I model myself and the way I will impact the observations I might see in the future. Those are actions, right? Those are actions. And if we in the basic picture, actions are just transition probability matrices on a state space. >> Yes. But how do I select one action over another? And >> you have an well I'm proposing uh a particular objective function that you see that you're seeking to >> Yes. Um and the the way in which this objective function works as you can see in this in this is exactly the same as it would work for any of the states as well that I don't control. Yes indeed I I I I agree that this this is a fascinating question you know about the difference between reflexive actions um and if you like deliberate actions if they somehow require uh different modeling I I I'm inclined to think that they don't but that's who who am I to because I'm coming at this as as an outsider. But if they don't require separate modeling, then I believe that the framework I've presented is adequate. >> Yes, I think on on that assumption, you know, with that caveat of saying predicting a a sort of passive playing out of of the world and of of the actions I can take in it, if there is no difference between those things, then I completely agree. It's >> well, is it Okay. If you could point me to a paper where this uh issue is uh trashed out, I I would be very keen to u to read it. >> When you say this issue, you you mean >> maybe whether uh there is a fundamental difference uh reflexive actions. Okay. And so each has to infer what he's doing. and um deliberate actions which uh involve uh offline some sort of distance from um what's going on where the where the where that is causally uh um efficacious. Uh >> yeah. >> Yes. And and you know I think this is actually one of those really interesting connections again back to what we were saying about the difference between the way a lot of the continuous space state space models work and the way in which a lot of the the uh discrete omdp like models are are formulated. Um, and the I guess the key thing in a lot of the way the continuous models are and this isn't necessarily because they're continuous, but that but they match the same sort of principles that you've been describing in that there is no need to explicitly model myself or or or maybe I I might model the position of my limb for instance and predict where I might see it at some point in time. Um but what I don't have to model necessarily is uh how I will make decisions, how I will plan. There's no sort of self-modeling involved in it. But as soon as you say how the world evolves depends upon which decisions I make, one needs to have prior about which decision they are likely to make given a set of beliefs that they would have at that point in time. And it's only by incorporating that that you can then predict futures that that that we ourselves influence and create. So that's that's the distinction. It it and in terms of the modeling, you know, you can still use exactly the same machinery you've described, but you need to change the prior. So again, it comes back to this point that the model is central the general. >> Yeah. The question in my mind is whether everything can be encoded in the HM. >> Well, I think it you know you you can certainly write down an HMM in which there are some states that are states and there are some states that are policies. and and I think that's what you're describing. But the question is, >> is there something meaningfully distinct between those two different kinds of entity? >> That's what that's what I'm sort of pushing back against. Okay. I I I I'm aware of this issue and um I I find it fascinating and I'm actually just putting my foot down and saying >> yeah, >> this is not a real problem. There's a sense in which in the total joint distribution of let's say agent and niche again it is great to have an evenhanded approach to those variables. At the same time it's like the difference between external context shifts which are changes in potentially causal external influences that there are uncertainty profiles over. And then there's the kind of internal context shifts or actions and the agent in the specification and the implementation. You have that agent doing actual agencyoriented policy search over what I could do. I could move my eyes there. I could look here and that has all these implications for precision and salience. Whereas you still could have uncertainty profiles and transition matrices over like I'm a marionette and my arm will be moved there by this external force. But that would be represented with an uncertainty over those those changes in the world which are not under one's own agency versus ones that are. And it's like you need to do policy inference for the true I could do that whereas this could happen to me you could just be uncertain about like whether it rains or not. Well, let me let me give the counterargument, okay, which which would say that um uh the agent's job is to predict its actions and the way it predicts its actions is by predicting its sensations. It's interceptive and auton or whatever proprioceptive uh sensations and then fulfilling those uh predictions because on sufficiently short uh time scales those predictions can be fulfilled uh reflexively motor reflexes and and autonomic reflexes. So if the agent can successfully predict its future sensations, successfully predict them, okay, then it can fulfill them um unconsciously as it were. I mean without without having to so that's why I think that the reflex of picture is sufficient. Yes, there is this sort of like ultimate witness view where everything including what might be seen as intentional or even internally perceived as intentional is also like a willless unfolding. That's I I guess I'm I I am the plastic determinist at heart and um >> that's that's how that's how I see nature including human nature and I see a lot of um push back against that in the act of inference literature or a lot of um shall we say humanists who who don't like this uh purely mechanical um statistical physical account of of human behavior, but it it it works for me. >> Well, I think they are perhaps slightly different distinctions. I I would argue that that sort of conventional accounts of active inference don't necessarily appeal to anything other than statistical and physical. Well, well, I mean there is the field has generated so much interest uh in the humanities. Okay. that um >> and that's because I think uh humanists see a bridge between um well car's uh worldview and and their own but my reading of Christian is that it is uh deeply mechanistic and um ultimately reflexive. I think >> yes and I you know I agree with you the reflexive aspect is incredibly important and I think a part of all active inference formulations um the the idea of it being of just having the and and you know the at the sort of lowest level of the model where we interface with sensory data I think that is always the assumption that simply to fulfill predictions and what you've described today is is precisely the philosophy and approach that underlies the the quite traditional continuous state space approaches to active inference. >> And there is no reason why one can't apply exactly those same approaches equally validly to to discrete state spaces. Um so you know the the approach you you you've outlined is completely reasonable. Um but it addresses a very different question and problem to that addressed by expected free energy. So it's a different domain. It's a it's a different category of thing. The expected free energy is a way of saying if I were to predict what I'm going to do next, knowing the beliefs I hold now, how would I weight those alternative actions, those alternative trajectories, those alternative choices in relation to the beliefs I have now. And that comes down to a question of prior. It's a question of um how do I believe I would behave in this situation? And one can't do that with a prior that is just a fixed form. I always do this in this situation which is is what you're describing in terms of >> well I I I think we we we are now um circling the same u problem you know as I see it the problem is uh is it possible to um formulate a single generative model such as HMN which encodes all of uh an agent 's uh beliefs um actions, policies uh its whole world cannot be encapsulated in a single HM and if it can then the calculations with that HMM will be just mechanistic. >> that's the question. Uh, of course I don't. Yeah. >> And and I think the point I'm making is that that your answer is a perfectly reasonable answer, but you're addressing a different question to what is expected free energy for. So you could have written this paper and this talk without reference to expected free energy. >> Oh, that's well that's interesting. >> And I would have had very few very few uh points of contention. >> Oh, okay. Well, I I would certainly certainly be uh curious to uh to hear of any other, you know, judgments u that other people might um have on on that uh on that particular issue. Um yeah, it could be that I've completely misunderstood it because I've just come as a complete outsider and um well, if I was doing it, then this is how I would do it. And uh that was what prompted me to prepare for it. >> Not not about being an insider or outsider. This is a right of passage. This is something that that sort of many people go through. I think is >> okay. >> Okay. That's um I think maybe we should uh we should conclude on that now for >> Yeah. It it is an evergreen genre. first principles from first principles and all of our journeys and our abilities as time goes forward to integrate these formalisms and come to more and more consolidated streamlined representations of them. It's just the beginning of that day. So Patrick, thank you for the paper and I hope it spurs correspondence and dialogue. And Thomas, thank you for the textbook and all the work and these comments here today. Thank you. Thank you. It was my pleasure. I I enjoyed the discussion very much. >> All right. Bye. Okay. Thank you. See you later.